

Paper 3 — Handoff Note v3

Isaac Sim tactile data-collection pipeline — pad-pose mismatch investigation

Supersedes Handoff v1 (Jul 3) and v2 (Jul 7). Written end of the Jul 9 debugging session.

How to use this note: read Section 1 for the one-line state, Section 6 for exactly what to do next. Sections 2–5 are the evidence trail so a fresh chat does not repeat work already done.

1. Project and current state

Project. Paper 3 replaces the hand-crafted shape-classifier + extrapolation rules of Papers 1–2 with a **learned contact-completion model**: input one tactile image from a single contact, output the full unseen contact map across the pads, so the Paper-2 virtual grasp search can run over a large learned tactile map. Data is generated in **NVIDIA Isaac Sim 5.1**: UR5e + Robotiq 2F-85 gripper + two TSF-85 tactile sensors, grasping a cylinder (26 mm dia, 140 mm long).

Four-block plan. Block 1 (Isaac factory) working. Block 2 (stitching) is the live front. Blocks 3–4 (train two U-Nets sim-vs-sim+real, then A/B) not started.

Current blocker (one line). Pads land ~25–30 mm too high vs the GUI-designed pad pose. Root cause is now understood; a clean physics measurement of the closed-grip pad pose is the last missing number, and it is blocked by an intermittent gripper-close NaN.

2. The pad-pose mismatch — what we proved

The vertical logic in `collect_from_config.py`: command the wrist (EE) to `GUI_pad_target_z + TOOL_OFFSET_Z`, so the pad ends up `TOOL_OFFSET_Z` below the wrist. Proven from `pad_truth_probe.json` + pose history:

- **The arm is flawless.** EE reached = commanded EE to within 0.08 mm on every grasp. Grid steps exact (+7.99 / +8.01 vs 8.00 mm commanded).
- **The mismatch is entirely one constant:** `TOOL_OFFSET_Z = 190.24 mm`. It overstates the true wrist→pad-face distance, so the wrist is dropped too high and the pad lands ~25–30 mm high.
- **It is NOT the finger-swing arc** (an early theory). The 190.24 was calibrated on this same 26 mm cylinder closed, so any swing at this diameter is already baked in. Direction was also backwards (pad high, not deep).
- **It is NOT bad finger-link mass** (a later theory). Mass probe: `right/left_inner_finger` have mass 0.0393 kg, all-positive inertia, proper `RigidBody` APIs. Healthy.

Correct model going forward (agreed): wrist target = GUI pad pose – (wrist→pad offset at that object's closure angle). One problem, two parts: the offset must be (1) correct in the first place, and (2) it changes as the gripper closes (finger swing), so it must be measured at the closed state we actually grasp in. Get the offset by **measuring** the fingertip pose in sim at closed hold — not from assumed CAD geometry, because this asset is a defeatured/modified Robotiq with a custom adapter and TSF-85 pads bolted on.

3. Bug fixed this session (already in the code you have)

pad_desired was garbage. Old collector computed `desired = INIT_PAD_POS + (dy, dz)`, but `INIT_PAD_POS` is the home pose while (dy,dz) are offsets from the object centre — wrong anchor. It faked a constant ~ (0, +150, +199) mm miss on every grasp. Now writes `pad_gui_target_m = object_centre + (0, dy, dz)` natively into `pose_history.json`. If any old plot showed the pad ~150–200 mm off, that was this, not the robot.

4. The current obstacle: intermittent gripper-close NaN

On the latest 1×1 probe run, during gripper close the finger link jumped to world (+838, −94, 1056) mm, `finger_joint_rad = NaN`, and `SingleRigidPrim` returned `NaN`. Isaac showed: *“Invalid PhysX transform detected for ... right_inner_finger.”* This is a physics instability during close, not loss of robot control.

What we know about the sensor prim: the probed `.../TSF_85_right/TSF_85` is an **empty Xform group** (no physics APIs). The real deformable body is one level deeper (a third nested `TSF_85`), and that deeper child is the source of the recurring *“possibly invalid inertia tensor / negative mass”* warning seen in every screenshot.

User's recollection (Jul 9): earlier grasp runs *did* close normally and produced real (large) tactile files. That points to the NaN being **intermittent** rather than a permanent broken asset — so the plan is a NaN-guard + retry, plus verifying the deeper deformable is what needs stabilizing only if retries keep failing.

5. How the robot is actually moved (for reference)

Arm motion = **cuRobo motion generation**, not hand-written IK/Jacobian. In `collect_from_config.py`:

- `plan_free_move` — `cuRobo MotionGen.plan_single` for the big hop to the approach point.
- `plan_stitched_z` — the Paper-2 incremental descent: splits the vertical move into `N_STEPS=10` small steps, holding tool orientation via `PoseCostMetric(hold_partial_pose=True, hold_vec_weight=[1,1,1,1,1,0])` (the zero frees one axis so the pad slides straight down).
- `fk` — `cuRobo` forward kinematics (joints → EE pose). Pad FK pose = EE FK + a FIXED wrist→pad constant, which is BLIND to the finger swing (that is why FK cannot measure the real pad).
- Gripper = `ramp_gripper` ramps `finger_joint` 0→0.55 rad over 60 frames via `apply_action`. This is the step that NaN'd — separate from the arm / `cuRobo`.

6. What to do next (pick up here)

Goal: get ONE clean closed-grip measurement of the real pad pose, then replace 190.24 with the measured wrist→pad distance.

- 1 **Re-run the 1×1 centered grasp** (`nx=1, ny=1, dy=0, dz=0`) once or twice. If close is clean (no NaN in `pad_truth_probe.json`), the `right_inner_finger` closed-grip world pose is the number we need. Compare it to `gui_target_m` to get the true offset.
- 2 **If NaN recurs:** add a NaN-guard + auto-retry in `grasp_one_point` (detect NaN `finger_joint` / non-finite pose, re-run that grasp). This was the agreed Fix Path A.
- 3 **If retries keep failing:** stabilize the deeper `TSF_85` deformable child (mass / inertia / solver settings) — Fix Path B. Only go here if A does not hold.
- 4 **Once a clean closed pose exists:** wire the measured value in as `pad_measured_pos_m`, set the corrected offset, and the pad-truth safety report flips from amber (FK-only) to green/red. Then re-verify the pad sits on the rod in the sim view.
- 5 **Then resume Block 2** (stitching), which was paused pending trustworthy poses.

7. Files, tools, and hard constraints

Files touched this session:

- `sim/collect_from_config.py` — fixed `pad_gui_target_m`; added pad-truth probe (pt00 only) reading 4 paths (right/left_inner_finger + both TSF prims) at open & closed grip, writing `pad_truth_probe.json`. Probe is fully wrapped, cannot break a run.

- viz/pad_truth.py — NEW. Safety report: GUI-designed vs reached pad pose per grasp; amber 'FK-only' banner until a physics measurement exists, then green/red at THRESH_MM=3.
- main_gui.py — added 'Check Pad Truth (safety)' button + show_pad_truth method.

Confirmed prim paths (note the DOUBLED base):

/World/robot_gripper_adapter_sensor/**robot_gripper_adapter_sensor**/Robotiq_2F_85_adapter_fixed_v_sibling__1_/Robotiq_2F_85_modified/Robotiq_2F_85/**right_inner_finger** (and left_inner_finger). Sensor prims: .../robot_gripper_adapter_sensor/TSF_85_right/TSF_85 (empty group; real deformable is one level deeper).

Hard constraints (carry forward):

- Isaac SWALLOWS print() in script mode — ALL probes/diagnostics MUST write to a file; file is ground truth.
- Read the TRUE pad/finger pose from the PHYSICS side (isaacsim.core readers / usdrt), never UsdGeom.XformCache (the frozen-pose lesson).
- s1 = right = raw; s2 = left = mirror L-R. Reshape (7,4), never transpose. SUBTRACT_BASELINE=False, margin=0, signed/anchored grid.
- Arm = cuRobo (curobo), not raw IK. Two-Python wall: Isaac side vs plotting/GUI side kept separate.
- Working style: ONE verifiable step at a time; measure, do not guess; short plain explanations; distances in mm; when Kourosh's eyes disagree with a diagnostic, the diagnostic has been wrong every time — instrument further.

8. Open questions for the next chat

- Does re-running produce a clean (no-NaN) close, giving a valid closed right_inner_finger pose?
- The finger LINK origin may sit a few mm from the actual pad FACE — once the measured closed pose is next to the GUI target, decide if a small face-vs-link correction is needed.
- GUI 3D panel: robot base is plotted but off the clamped axes; and the default 3D camera mirrors X vs the FRONT panel. Cosmetic fixes proposed (auto-fit limits or edge arrow; view_init(elev=22, azim=-90)); not yet applied.
- A-vs-B grid-tilt: when the object tilts, should the grasp grid rotate with it? GUI currently builds an upright grid (no tilt rotation). Still to decide.

End of Handoff v3. The single most important next action: re-run the 1×1 centered grasp and capture a clean pad_truth_probe.json, then read the closed-grip right_inner_finger pose.